

INVESTOPIA



Team Name: *FireLinx*

Business Type: Product-based Service

Industry: Fire Protection & Safety-Technology

Fire Intervention &
Rapid Extinguishment over Long-
Distance IoT-Network Execution



PROBLEM STATEMENT

OVERVIEW OF THE PROBLEM:

⚠️ Response & Containment Challenges:

- **Sprinklers & Smoke sensors:** Fail upon large-scale fire
- **Worsened Fire-Brigade Response:** Delayed alerts, traffic, and narrow road - accessibility
- **Rapid Fire spread:** Late Response leads to Prolonged & Increased suppression efforts

⚙️ Infrastructure & Systemic Limitations:

- Existing systems lack **Sensor abundance** & Intelligence
- Infrastructure is **less-flexible** & structure-invasive
- **Manual intervention** endangers both rescuers and trapped
- Remote Oversight Problem over Geographically Challenging Location

INDIA'S FIRE CRISIS IN NUMBERS:

18000+
deaths per
year

40% slums
ignite
without
safety

24,000
forest fires
rage per
year

20,000+
industrial
incidents
safety gap.

WHY MINUTES MATTER !!

Traditional firefighting faces deadly delays:

- *Dispatch → Travel → Setup = Lost time.*
- *Sprinklers fail. Hoses can't reach. Hydrants are inaccessible.*
- *High-rises, slums, forests, factories = logistical nightmares.*

SOLUTION

Our device FireLinx is a Multi-modal Instantaneous automated Fire-Suppression system, which bridges these gap :

Adaptive Suppression:

- Multi retardant Technology
- Dynamic Pump Speed

Remote Oversight & Control:

- Web-Dashboard & telegram-Bot
- Secured Authoritarian & Public Access

Resilient Communication:

- 24/7 Online Monitoring System
- 6 Channel Failproof Communication

Intelligent Detection & Targeting:

- Real-time Edge-Computing
- Multi-Sensor Array System
- Physics Calculation: Haversine & Bearing Formula for Precise Target



Fig. 1. FireLinx: Mai Station STAFERB



Fig. 2. Telegram Bot Fire-Alert Data

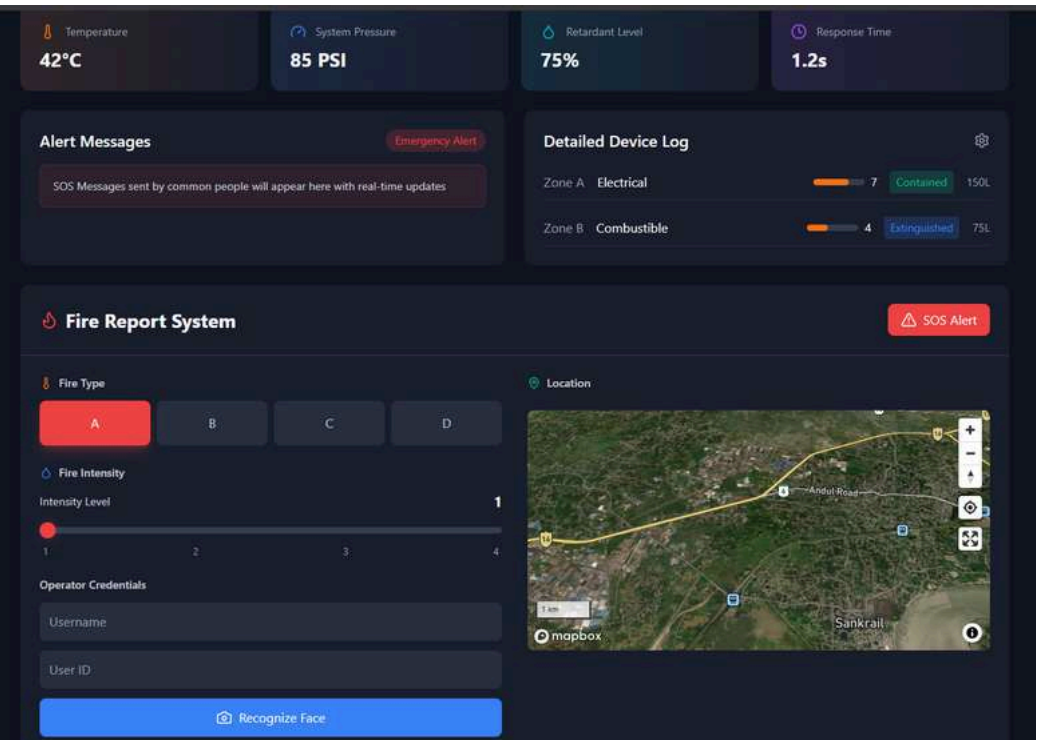
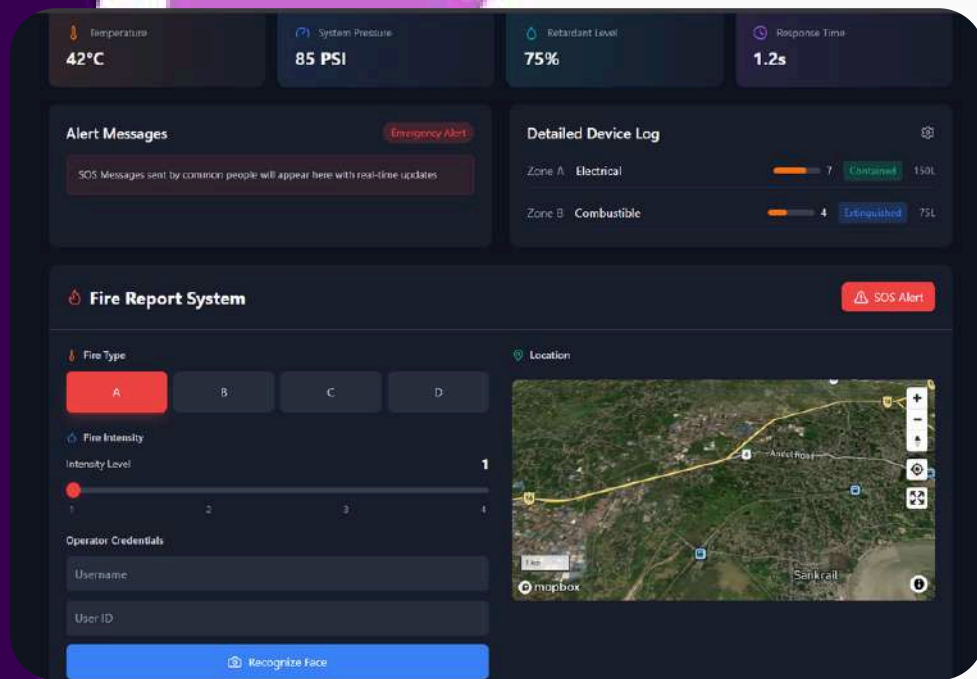


Fig. 3. Web-Dashboard for Fire-Alert Data

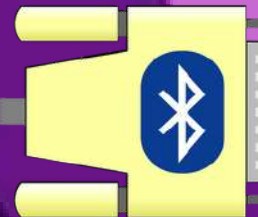
PRODUCT MVP



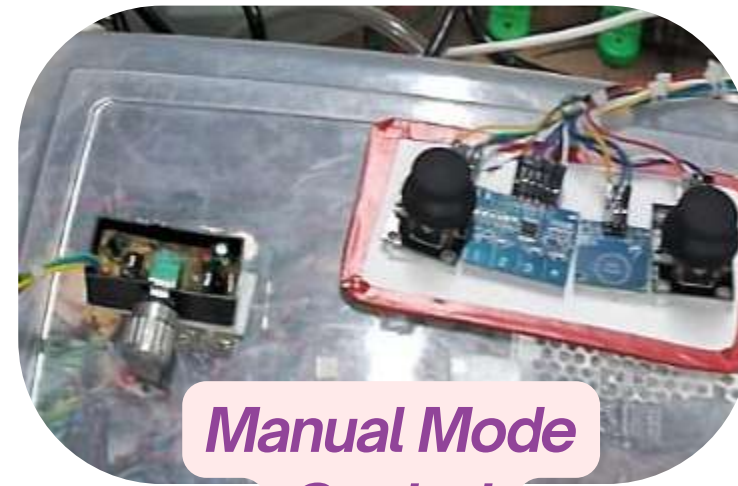
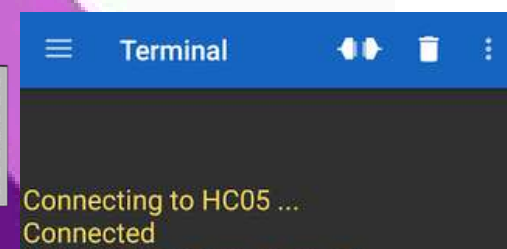
Firelinx Online Web-Alert Dashboard



Telegram Bot for Local people to Send Alert



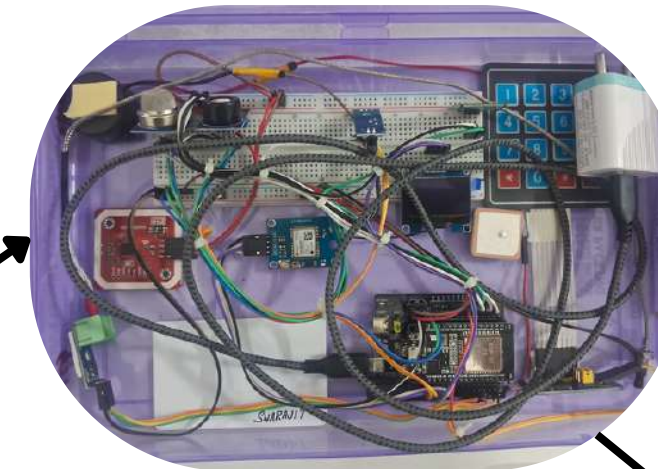
Bluetooth Serial communication



Manual Mode Control



Main Station: STAFERB



ESP32-Subsystem

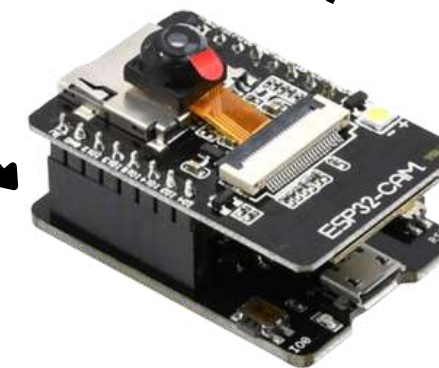


Manual Fire Alert

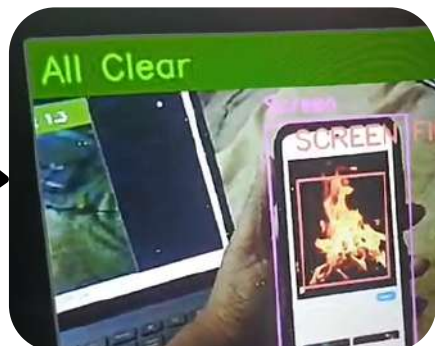
Sensor AUTO Fire Alert



3-Step Fire detection ML model



ESP32-CAM Subsystem

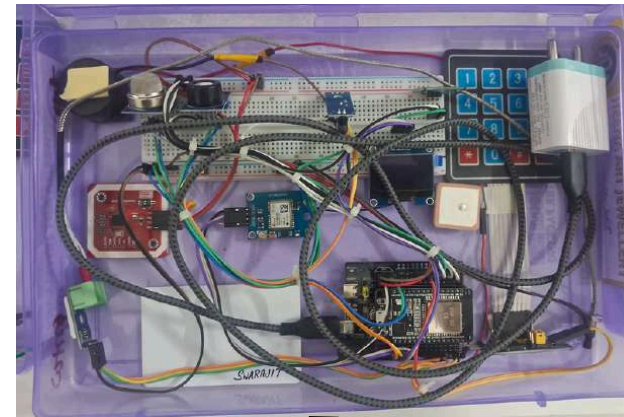


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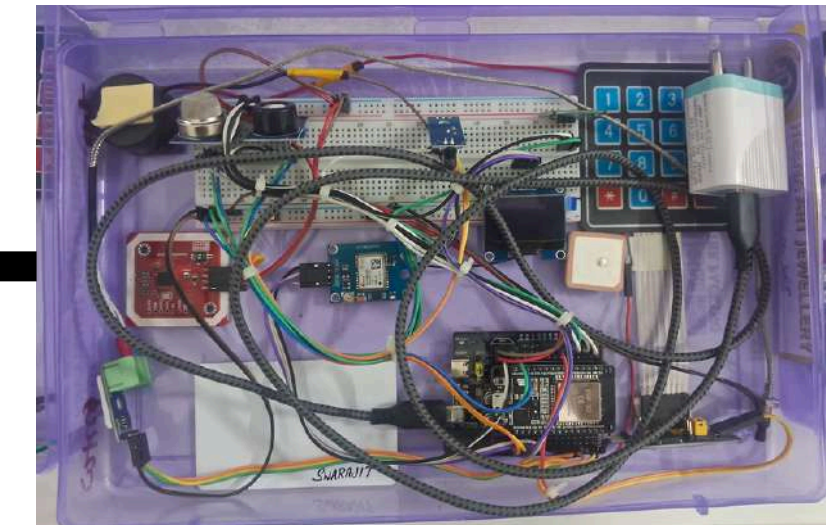
PRODUCT MVP

On-Site Distribution
of Physical Unit at
various zones

ESP32-Subsystem
StnID: A/1 (North zone)



Main Station:
STAFERB

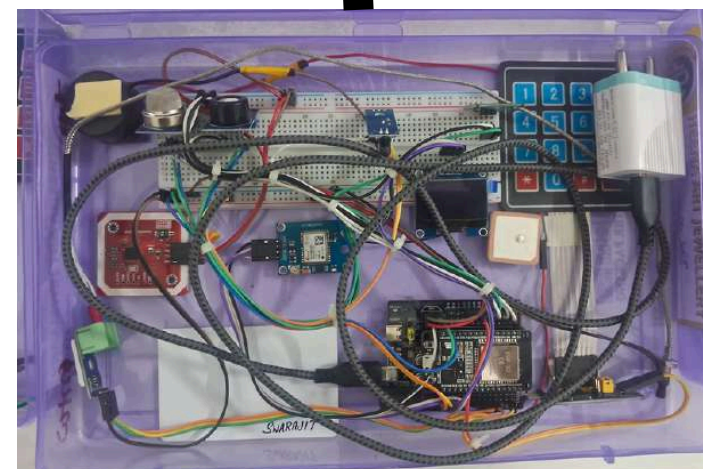


ESP32-Subsystem
StnID: B/2 (East zone)

ESP32-CAM Subsystem
StnID: E/5
(South-East zone)



ESP32-Subsystem
StnID: C/3 (South zone)



ESP32-Subsystem
StnID: D/4 (West zone)



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MARKETING PLAN

Initial Market Launch & Growth Strategy (24 Months)

- Factories
- Warehouse
- Fuel Depot

Tier 1:
Essential
Guardian

- 3x Guardian (Std) Station;
- 12x Sentinel Nodes

Objective

- Align with *Corporate Capex Cycles*
- Secure *High-Margin* Recurring *Revenue*
- Target *Immediate Profitability*



- Industrial
- High Rise
- Residential

Tier 2:
Fortress
Grid

- 2x Guardian (Std) Station;
- 16x Sentinel Nodes

Objective

- *Overcome* Community *Funding Hurdles*
- Secure *High-Margin* Recurring *Revenue*
- Target *Immediate Profitability*

- Municipality
- Slum Area
- Hilly Forest

Tier 3:
Public
Shield

- 1x Guardian (Big) Station;
- 20x Sentinel Nodes

Objective

- Anchor growth with *Large Contracts*
- Align with *National Safety Mandates*
- Establish as a *Key Public Vendor*



UNIT ECONOMICS



Our go-to-market strategy is centered on offering flexible, scalable safety ecosystems, not just a single product. This allows us to address the unique needs and budgets of diverse client profiles, from small workshops to large-scale public infrastructure.

Table 1: Summary of Bespoke FireLinx Solution Packages

Package Tier	Target Client	Typical Configuration	Hardware Price	Annual AMC	Total Year 1 Price
Tier 1: Essential Guardian	• Small-to-Medium Factory, • Warehouse, • Fuel Depot	• 3x Guardian (Std) Station; • 12x Sentinel Nodes	₹5,00,000	₹2,70,000	₹7,70,000
Tier 2: Fortress Grid	• Large Industrial Plant, • High-Rise Building Complexes • Residential/Commercial	• 2x Guardian (Std) Stations; • 16x Sentinel Nodes	₹3,50,000	₹2,60,000	₹6,10,000
Tier 3: Public Shield	• Public & Municipal Spaces, • Dense Slum Area, • Forest Periphery	• 1x Guardian (Large-Area) • 20x Sentinel Nodes	₹4,30,000	₹2,50,000	₹6,80,000

COMPETITOR ANALYSIS

Long term Cost Comparision of Legacy vs. FireLinx Solution

Client Profile	FireLinx Solution	Legacy Solution	Upfront & 5yr Term Cost		Fire Loss	Loss Averted		Client's ROI	
			FireLinx	Legacy		FireLinx	Legacy	FireLinx	Legacy
Heavy Industry / Warehouse	Essential Guardian: (3 Stations, 12 Nodes)	Industrial Sprinklers & Hydrants	₹18.5 L (=₹5L + ₹2.7L x 5yrs)	₹19.5 L (=₹12L + ₹1.5L x 5yrs)	₹1 Cr	₹95L	₹15L	413.50%	Nominal (25%)
Residential Colony / RWA	Fortress Grid: (2 Stations, 16 Nodes)	Basic Compliance (Alarms, Extinguishers)	₹16.5 L (=₹3.5L + ₹2.6L x 5yrs)	₹7.5 L (=₹5L + ₹0.5L x 5yrs)	₹75L	₹71L	₹4L	354.55%	Negative (-20%)
Municipal / Public Space	Public Shield: (1 Big Station, 20 Nodes)	Limited Hydrant Access	₹16.5 L (=₹4.3L + ₹2.5L x 5yrs)	₹5 L (=₹3L + ₹0.4L x 5yrs)	₹50L	₹47L	₹1L	184.85%	Negative (-67%)

COMPETITOR ANALYSIS

Water Sprinklers

- ✓ Best for indoor fire
- ✗ Not for big fire
- ✗ Not Intelligent



Drone

- ✓ Mobility
- ✓ Sensing Capability
- ✗ Retardant carrying capacity
- ✗ Delicate Equipment
- ✗ Weather Restriction



Fire Hydrant

- ✓ Immediate ,reliable water
- ✗ Invasive to building structure



Manual Fire Truck

- ✓ Well equipped
- ✓ Courageous
- ✗ City traffic
- ✗ High response time
- ✗ Road blockage



FireLinx

- ✓ Suitable for big fire
- ✓ Sensor based Intelligence
- ✓ Has 3 types of redardants
- ✓ (CO2/Foam/Water)
- ✓ Not so delicate equipment
- ✓ No weather restriction
- ✓ Not invasive to building structure
- ✓ Invariant to city traffic
- ✓ Very less response time.
- ✓ No effect of road blockage.
- ✓ Can be installed in urban slums.



UNIQUE SELLING POINT

Integration with Existing Infra:

- Linked to central water tanks
- Ensures strong, sustained response



True Multi-Hazard Suppression:

- Supports water, foam & CO₂
- Works for all fire types.



Serves all kinds of regions:



Dense urban slums



Industrial facilities



Remote forests

Customization:

- Flexible, customizable node setup
- Scalable sensing and response stations



Planned Complexes



School/Hospital



Railway Station

Monitoring & security :

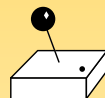
- All-in-one 24/7 smart monitoring
- RFID cards given to Authorised Personnel



Multi-Channel Communication:

Alert is sent via:

1. Manual mode



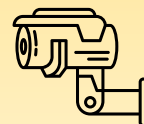
2. Sensor detection



3. Web Dash



4. Camera Subsystem



5. Telegram Bot



Telegram

6. Bluetooth channel



Municipal Area



Warehouse



Market Complexes

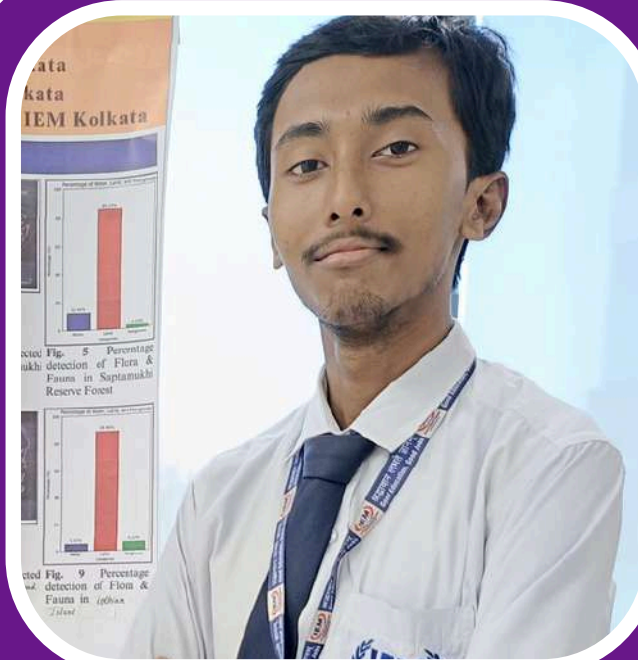
TEAM DETAILS



Adrija Ghosh



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Apratim Haldar

TEAM

FireLinx



Other Members:

- 1.Md. Mustafa Abdulla**
- 2.Swarajit dey**
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[Github Link](#)



[Pre-build Demo Video Link](#)

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